A Contextual Embedding Analysis of Four Chinese Internet Terms

Research report by Xinbei Mi

Abstract

This report examines four Chinese internet terms—绷 (bēng), 哈 (hā), 拉 (lā), and 塌 (tā)—using Bilibili comment and crawl records, contextual embeddings, exploratory clustering, and descriptive temporal summaries. The project uses hfl/chinese-roberta-wwm-ext, a Chinese whole-word-masking RoBERTa-family encoder. It extracts contextual representations from one fixed pretrained model, compares masked-target and target-token representations, and applies density-based clustering with refinement in selected analyses. The figures show distinct patterns across the four terms: 绷 includes physical, restraint, evaluative, psychological, and residual uses; 哈 is dominated by laughter-like forms, with smaller 哈基米-like, breath-related, and 二哈 groups; 拉 spans physical and internet-associated forms; and 塌 includes physical-collapse, fandom-related 塌房, idiomatic, and other uses. The results are descriptive snapshots, not a validated history of Chinese internet language. Uneven samples, exploratory labels, and differences between analysis rounds limit temporal claims. These case studies examine how contextual embeddings can help describe emerging Chinese internet slang; they do not constitute a direct test of whether a pretrained model understands slang.

Keywords: contextual embeddings; lexical semantic change; Chinese internet language; word sense induction; semantic analysis

1. Introduction

Internet expressions often develop new pragmatic uses through repeated participation in platform-specific contexts. A form may retain a literal use while acquiring evaluative, interpersonal, or community-specific functions. Research on Chinese new-media vocabulary has described semantic extension and grammatical change as tendencies toward more interactional uses, while contextual collapse and social-network conditions are proposed as motivations (He & Liu, 2024). Liu, Liang, and Wan (2023) similarly discuss "interactivization" in the semantic and grammatical development of novel lexical usages. These accounts motivate asking how a word's contextual neighborhoods vary across time, but they do not establish that every internet term follows the same trajectory.

The project presentation also draws on cognitive approaches to metaphor and embodied meaning, and on the lecture's discussion of form–meaning associations, contextualized language models, and lexical semantic research (Liu & Qing, 2026; Lakoff & Johnson, 1980). These ideas serve here as interpretive lenses. For example, a movement or bodily-state expression may be used in a social or evaluative context; an embedding cluster can indicate recurring contextual similarity, but it cannot by itself demonstrate a metaphorical

mapping, a pragmatic function, or a historical pathway. Such interpretations require examination of actual concordance lines and independent linguistic coding.

The study asks three descriptive questions: (1) What recurring contextual or form-based use groups appear for 绷, 哈, 拉, and 塌 in the available Bilibili corpus? (2) How do their observed proportions vary across recorded years? (3) How closely do contextual embedding clusters align with project-assigned use categories? The analysis is exploratory. It reports project snapshots and their denominators rather than treating outputs from different experimental rounds as one harmonized dataset.

2. Data and Common Method

2.1 Corpus and preprocessing

The project collects Bilibili material through video discovery and comment/danmaku collection scripts. The publicly prepared repository is code-focused; raw and processed user-generated data, credentials, model weights, and project output files are excluded. The counts in this report come from local handover notes, logs, and result snapshots, not from a newly rerun end-to-end pipeline. Records were normalized, target-word occurrences were located, obvious noise and non-target uses were filtered, and near-duplicates or repeated records were handled in the cleaning workflow. Usernames and identifying fields are not used as semantic evidence. The precise filters and denominators vary by target and by analysis round; the individual experiment sections state the available counts. The collection is concentrated from 2016 to 2026.

2.2 Model and representations

The model named in the code and project notes is hfl/chinese-roberta-wwm-ext, a 768-dimensional Chinese encoder in the RoBERTa/whole-word-masking family (Cui et al., 2020, 2021; Liu et al., 2019). The weights are retrieved from Hugging Face and stored in the local cache, typically configured on the C: drive, rather than committed to GitHub. The pipeline extracts contextual vectors by averaging the final four hidden layers. It initially compares target-token and masked-target context representations. Because the present analysis focuses on contextual usage rather than direct lexical identity, the results shown here use the masked-target representation for clustering. This is an exploratory representation choice, not a benchmark evaluation of model understanding.

The project applies centering and an all-but-the-top (ABTT) operation to reduce dominant common directions in embedding space, followed by normalization where required. These transformations can change neighborhood geometry, so clustering results depend on the representation and preprocessing choices. PCA is used for two-dimensional visualization only.

2.3 Clustering and temporal summaries

The analytical scripts implement an exploratory density-peak clustering workflow based on Rodriguez and Laio (2014), with density-chain assignment and cosine k-medoids refinement in selected pipelines. The clustering plots in this report follow the outputs of the project scripts; they should not be described as CLARA

results. The theoretical framing of contextual models in lexical-semantic research is also informed by Liu and Qing (2026)'s lecture slides.

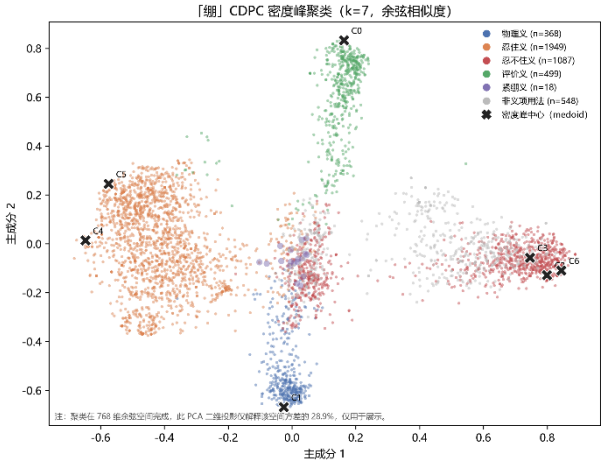
3. Experiments

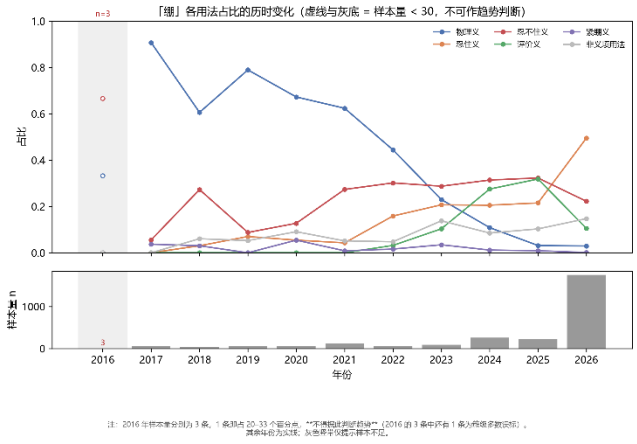
3.1 绷 (beng)

The 绷 analysis is the most fully documented case in the handover notes. The recorded pipeline begins with 4,767 target hits, retains 4,509 cleaned records, produces 4,488 vectors, and uses 4,469 records for the reported typology after excluding 19 rare or unassignable items. A narrower set of 2,701 records is marked canonical and judgeable for parts of the analysis. These denominators answer different questions and should not be interchanged.

The selected analysis used masked contextual vectors and k = 7 after reviewing candidate partitions. The notes report that an automatic k = 4 solution merged distinctions the researchers wanted to inspect, while k = 7 was retained as the smallest solution supporting the observed form/use basins. This is an analytic choice, not evidence that 绷 has exactly seven lexical senses. PCA's first two dimensions account for 28.9% of variance in the reported visualization.

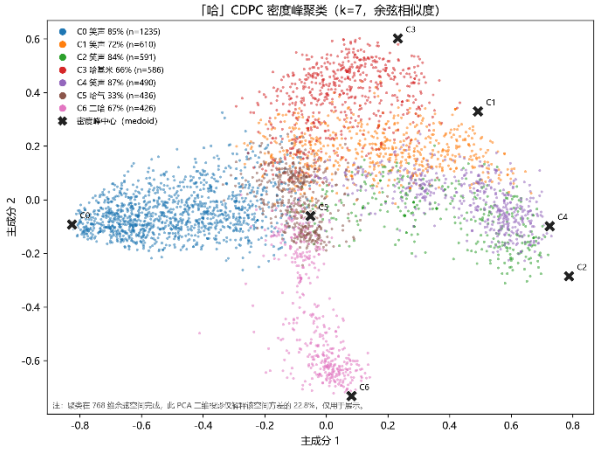
The final typology snapshot reports physical bandage/strapping use (n = 368; 8.2%), restraint or holding back (n = 1,949; 43.6%), inability to hold back (n = 1,087; 24.3%), evaluative 难绷 use (n = 499; 11.2%), tense or psychological use (n = 18; 0.4%), and non-use/residual items (n = 548; 12.3%). The 18-item psychological group is too small for a robust temporal frequency claim. One residual cluster had a reported purity of 0.49, indicating substantial category mixing.

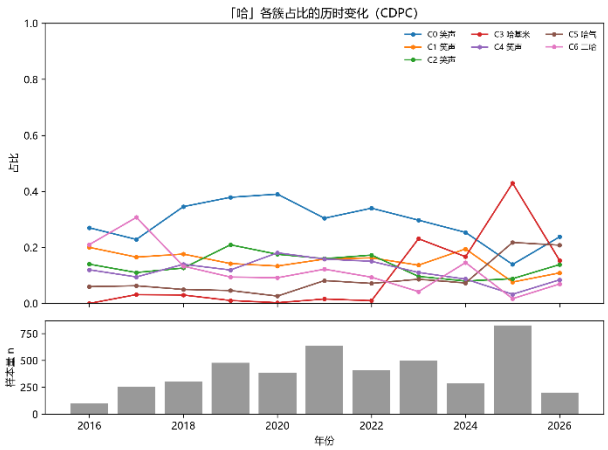




3.2 哈 (ha)

The latest reviewed ha snapshot records 4,503 cleaned items, 4,417 embedded records, and 4,374 clustered items in result3_final. It uses k = 7. Four clusters are primarily laughter-like by the project's form-based interpretation; together, laughter-form hits account for 2,502 of 4,374 clustered items (57.2%). Cluster-level form purities reported in the output are 0.845, 0.716, 0.843, and 0.869 for those four groups. The remaining clusters are associated with 哈基米-like forms (n = 586; purity 0.655), 哈气/breath forms (n = 436; purity 0.335), and 二哈 forms (n = 426; purity 0.669).

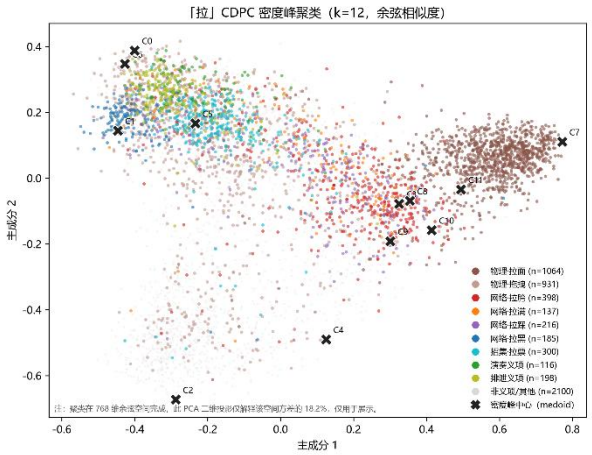


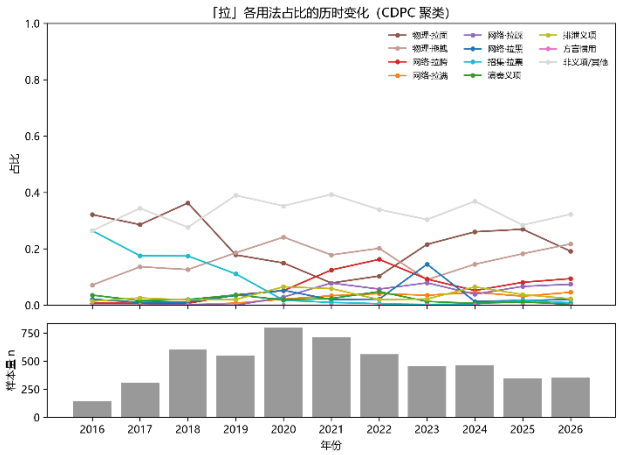


The 哈 results are exploratory rather than a fully adjudicated sense inventory. In particular, the low-purity breath-related group and polysemous or homophonous forms such as 哈基米 require contextual inspection before claims about borrowing, reference, or semantic development can be made. Yearly embedded counts range from 102 in 2016 to 834 in 2025 in the reviewed snapshot, with 2026 represented by 202 records; uneven yearly volume makes raw trend comparisons unsafe.

3.3 拉 (la)

The included 拉 figures show annual category shares and a k = 12 cosine-similarity clustering plot. The plotted cluster sample contains 5,645 records. Its legend groups the records into ten use categories: physical 拉面 (n = 1,064), physical 拉拽 (n = 931), online 拉胯 (n = 398), 拉满 (n = 137), 拉踩 (n = 216), 拉黑 (n = 185), 拉票 (n = 300), performance-related 演奏 (n = 116), 排泄 (n = 198), and non-target/other (n = 2,100). These figures are the basis for the description here; counts from other experimental rounds are not combined with them. The large residual category and form-based labels make the clustering exploratory rather than a validated sense inventory.



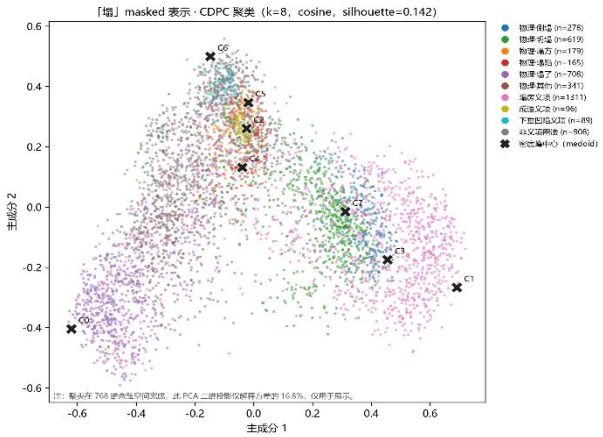


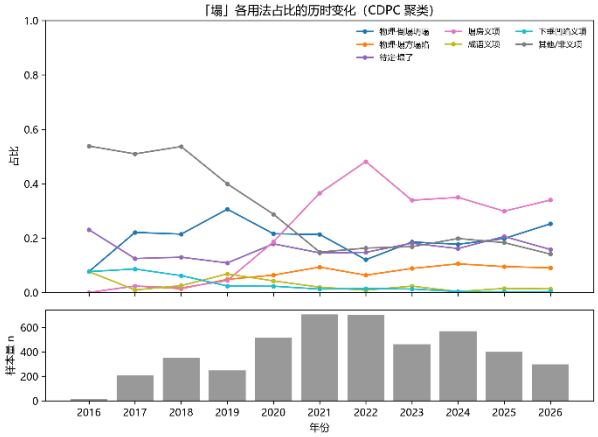
The annual-share plot shows variation in the relative proportions of the plotted 拉 categories from 2016 to 2026, with yearly sample counts displayed beneath the line chart. Because collection volumes differ across years and the plot is descriptive, the visible changes should be treated as leads for concordance analysis rather than evidence of a population-wide semantic shift.

3.4 塌 (ta)

The included 塌 figures show a masked-vector, k = 8 cosine clustering plot with a silhouette score of 0.142 and 4,694 plotted points. To match the figure, this section reports its legend counts: physical-use groups (n = 278, 619, 179, 165, 708, and 341), 塌房 (n = 1,311), idiomatic use (n = 96), face-related use (n = 89), and non-use (n = 908). A separate result4 typology table contains 4,691 items; it is a different snapshot and is not combined with the plotted count. The project also records 115 人设崩塌 occurrences incorrectly assigned to a physical-collapse category, so physical-use labels should be interpreted cautiously. The temporal figure shows substantial variation in yearly sample counts, including a very small 2016 sample; those proportions do not establish a historical trend.

The temporal summary shows a larger share of 塌房 in some years around and after 2021, but the year counts and sampling choices are uneven; 2016 contributes only a small number of analysis records. The handover also identifies an unresolved rule error: 115 occurrences of 人设崩塌 were assigned to a physical-collapse category. This known error weakens category-level conclusions until repaired and the downstream tables are regenerated.





4. Results

Across the four targets, contextual representations expose both recurring neighborhoods and substantial ambiguity. The project's strongest descriptive signal is the dominance of laughter-like forms for 哈 in the latest cluster snapshot and the large restraint/holding-back groups in the 绷 typology. 拉 and 塌 show diverse physical, idiomatic, evaluative, and internet-associated forms, but their large residual or mixed categories show that automatic grouping and current category rules do not fully resolve usage.

Target	Latest reviewed snapshot	Main descriptive result	Caution
绷 beng	4,469 typology items; k = 7	Restraint / hold back 43.6%; cannot hold back 24.3%	Psychological group n = 18; residual purity 0.49
哈 ha	4,374 clustered; k = 7	Laughter-form hits 57.2%	Exploratory labels; breath cluster purity 0.335
拉 la	7,906 category rows (result9)	Residual/other 38.49%; diverse form-coded categories	8,068 embedding metadata rows; denominator gap unresolved
塌 ta	4,691 typology items; later comparison k = 8	塌房 n = 1,311; mixed physical/online inventory	Known 115 人设崩塌 misclassification; silhouette modest

The visualizations in the project—cluster scatter plots, temporal-change graphs, heatmaps, and frequency plots—are useful for locating patterns and candidate contexts. They do not independently establish a semantic change. In particular, PCA plots compress much of the vector variance, clusters can reflect topic or repeated phrase templates, and year-wise proportions can change when collection or filtering changes.

5. Conclusion

This project provides a code-based exploratory workflow for comparing the contexts of four Chinese internet terms with contextual embeddings and clustering. Its results suggest that the four forms have markedly different distributions in the collected Bilibili records: 绷 includes restraint, evaluative, and physical uses; 哈 is dominated by laughter-like forms with smaller specialized groups; 拉 has a broad and partly unresolved usage inventory; and 塌 includes both physical-collapse and online/fandom-associated forms. The current evidence supports describing corpus patterns and generating hypotheses. It does not support a fully validated semantic history or broad claims about all Chinese social media.

6. Limitations

First, Bilibili is a single platform whose users, genres, and interaction formats differ from Weibo and other services. Results cannot be generalized to Chinese internet language as a whole. Second, the data are keyword-conditioned and uneven across years; recent years dominate some targets, some early-year cells are very small, and 2026 is incomplete. Temporal changes may therefore reflect retrieval and platform activity as well as language.

Third, target and masked contextual embeddings encode contexts but do not directly encode historical time. A fixed model may also carry training-data and domain biases, and contextual vectors can mix lexical sense with contextual variation (Kutuzov et al., 2022).

Fourth, cluster count selection, representation choice, density parameters, and rule-based labeling affect outputs. Several result folders are successive experimental rounds rather than a single reproducible final snapshot.

Fifth, category assignments are not uniformly supported by independent manual annotation; purity is measured against project labels and can inherit label errors. The known 塌 misclassification and small 绷 psychological group illustrate this risk.

7. Statement on AI Use

Codex and DeepSeek Harness were used primarily to assist with code organization, code drafting or modification, debugging, and workflow documentation during the project. This report was prepared by reviewing the project's local README, handover notes, scripts, logs, result tables, and presentation materials. AI tools were not treated as primary linguistic evidence or bibliographic authority.

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